Simple Calculator with Logger

1.0

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Chapter 1

Simple Calculator with Logger

1.1 Introduction

This small C++ project demonstrates how to document a multi-file application using Doxygen. It contains a Logger and a CalculationEngine.

1.2 Usage

Run the program to calculate a number raised to a non-negative integer exponent. The Logger prints status messages.

Chapter 2

Todo List

Member CalculationEngine::power (double base, int exponent) const

Add support for negative exponents and more operations.

Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

CalculationEngine
: Performs mathematical calculations and logs their results 9

Logger
: Prints messages with their corresponding log level 11

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

CalculationEngine.cpp
: Implements the CalculationEngine class 13

CalculationEngine.h
: Declares a simple mathematical calculation engine 13

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: Implements the Logger class 14

Logger.h
: Declares a simple logging utility 14

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: Entry point for the Simple Calculator with Logger example 15

Chapter 5

Class Documentation

5.1 CalculationEngine Class Reference

Performs mathematical calculations and logs their results.

`#include <CalculationEngine.h>`

Public Member Functions

- CalculationEngine (const Logger &logger)

  Creates a calculation engine.
- double power (double base, int exponent) const

  Calculates base raised to exponent.

Private Attributes

- const Logger & logger_

  Logger used by this calculation engine.

5.1.1 Detailed Description

Performs mathematical calculations and logs their results.

The CalculationEngine uses a Logger object to report when a calculation starts and to display its result. This example provides a power operation.

5.1.2 Constructor & Destructor Documentation

5.1.2.1 CalculationEngine()

```
CalculationEngine::CalculationEngine (
            const Logger & logger)  [explicit]
```

Creates a calculation engine.

Constructs a CalculationEngine.

Parameters

logger	Logger used to report calculation information.
logger	Logger used to report operations.

5.1.3 Member Function Documentation

5.1.3.1 power()

```
double CalculationEngine::power (
            double base,
            int exponent) const
```

Calculates base raised to exponent.

Computes the power using repeated multiplication.

Parameters

base	The number being raised to a power.
exponent	The non-negative integer exponent.

Returns

The result of base raised to exponent.

Todo Add support for negative exponents and more operations.

Parameters

base	The number being raised.
exponent	A non-negative integer exponent.

Returns

The computed power.

5.1.4 Member Data Documentation

5.1.4.1 logger_

```
const Logger& CalculationEngine::logger_  [private]
```

Logger used by this calculation engine.

The documentation for this class was generated from the following files:

- CalculationEngine.h
- CalculationEngine.cpp

5.2 Logger Class Reference

Prints messages with their corresponding log level.

```
#include <Logger.h>
```

Public Member Functions

- Logger (std::ostream &output=std::cout)
  Constructs a Logger object.
- void log (LogLevel level, const std::string &message) const
  Writes a message with a level label.

Private Attributes

- std::ostream & output_
  Stream used to print log messages.

5.2.1 Detailed Description

Prints messages with their corresponding log level.

Logger provides a small, reusable way to display messages. It is used by CalculationEngine to report calculations.

5.2.2 Constructor & Destructor Documentation

5.2.2.1 Logger()

```
Logger::Logger (
            std::ostream & output = std::cout)  [explicit]
```

Constructs a Logger object.

Constructs a Logger using the supplied output stream.

Parameters

output	The output stream where messages will be written. Defaults to the standard output stream.
output	Stream where log messages will be written.

5.2.3 Member Function Documentation

5.2.3.1 log()

```
void Logger::log (
            LogLevel level,
            const std::string & message) const
```

Writes a message with a level label.

Prints a message preceded by its log level.

Parameters

level	The severity level of the message.
message	The message text to display.
level	Severity of the message.
message	Text to print.

5.2.4 Member Data Documentation

5.2.4.1 output_

```
std::ostream& Logger::output_  [private]
```

Stream used to print log messages.

The documentation for this class was generated from the following files:

- Logger.h
- Logger.cpp

Chapter 6

File Documentation

6.1 CalculationEngine.cpp File Reference

Implements the CalculationEngine class.

```
#include "CalculationEngine.h"
```

6.1.1 Detailed Description

Implements the CalculationEngine class.

6.2 CalculationEngine.h File Reference

Declares a simple mathematical calculation engine.

```
#include "Logger.h"
```

Classes

- class CalculationEngine

  Performs mathematical calculations and logs their results.

6.2.1 Detailed Description

Declares a simple mathematical calculation engine.

6.3 CalculationEngine.h

Go to the documentation of this file.

```
00001
00005
00006 #ifndef CALCULATION_ENGINE_H
00007 #define CALCULATION_ENGINE_H
00008
00009 #include "Logger.h"
00010
00020 class CalculationEngine {
00021 public:
00027     explicit CalculationEngine(const Logger& logger);
00028
00038     double power(double base, int exponent) const;
00039
00040 private:
00042     const Logger& logger_;
00043 };
00044
00045 #endif // CALCULATION_ENGINE_H
```

6.4 Logger.cpp File Reference

Implements the Logger class.

```
#include "Logger.h"
```

6.4.1 Detailed Description

Implements the Logger class.

6.5 Logger.h File Reference

Declares a simple logging utility.

```
#include <iostream>
#include <string>
```

Classes

- class Logger

  Prints messages with their corresponding log level.

Enumerations

- enum class LogLevel { INFO , WARNING , ERROR }

  Represents the severity of a log message.

6.5.1 Detailed Description

Declares a simple logging utility.

This file contains the LogLevel enumeration and the Logger class. The Logger class prints messages with a level label.

6.5.2 Enumeration Type Documentation

6.5.2.1 LogLevel

```
enum class LogLevel  [strong]
```

Represents the severity of a log message.

The values help identify whether a message is informational, a warning, or an error.

Enumerator

INFO	General information about program execution.
WARNING	A situation that may need attention.
ERROR	An operation failed or encountered a problem.

6.6 Logger.h

Go to the documentation of this file.

```
00001
00008
00009 #ifndef LOGGER_H
00010 #define LOGGER_H
00011
00012 #include <iostream>
00013 #include <string>
00014
00022 enum class LogLevel {
00024     INFO,
00025
00027     WARNING,
00028
00030     ERROR
00031 };
00032
00040 class Logger {
00041 public:
00048     explicit Logger(std::ostream& output = std::cout);
00049
00056     void log(LogLevel level, const std::string& message) const;
00057
00058 private:
00060     std::ostream& output_;
00061 };
00062
00063 #endif // LOGGER_H
```

6.7 main.cpp File Reference

Entry point for the Simple Calculator with Logger example.

```
#include "CalculationEngine.h"
#include "Logger.h"
#include <iostream>
```

Functions

- int main ()

  Runs the calculator example.

6.7.1 Detailed Description

Entry point for the Simple Calculator with Logger example.

6.7.2 Function Documentation

6.7.2.1 main()

```
int main ()
```

Runs the calculator example.

Returns

0 when the program finishes successfully.

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